

1) What is Javascript?

JS is a high level, lightweight, dynamic programming language used to create interactive and dynamic web pages. It runs in web browser and can also run on servers using Node.js.

2) Why was JS created?

It was created by Brendan Eich in 1995 at Netscape. Its purpose was to make web pages interactive by allowing actions such as form validation, animation and dynamic content updates without reloading the page.

3) is Javascript interpreted or compiled?

JS is both

Earlier, JS was purely interpreted but in modern JS engine uses Just-In-Time (JIT) compilation, which compiles code into machine code before execution for better performance.

4) what are features of JS?

⇒ Light weight & fast

⇒ Dynamic typing

⇒ Object oriented

⇒ prototype based inheritance

⇒ Event driven

⇒ cross platform

⇒ supports asynchronous programming

⇒ First class functions.

⇒ single-threaded execution

5) why is JS single threaded?

JS uses a single call stack, it executes one task at a time in a single thread.

it handles asynchronous using, callbacks, promises, ASync/Await, Event loop.

6) Java vs JS

JS	Java
* Scripting language	* programming language
* Dynamically typed	* Statically typed
* Runs in browsers and node.js	* Runs on JVM
* Prototype-based inheritance	* class based - inheritance
* Used for web development	* used for enterprise, desktop, android apps.
* Interpreted / JIT compiled.	* Compiled to byte code

→ what is ECMAScript

ECMAScript (ES) is the official standard specification for JS maintained by ECMA International.

Eg: ES5, ES6

8) What is role of Javascript engine?

Used for,

- ⇒ Reads JS code
- ⇒ Parses the code
- ⇒ Converts it into machine code
- ⇒ Executes the code

eg: V8, SpiderMonkey, JavaScriptCore.

9) What is V8 Engine?

V8 is Google's open source JavaScript engine developed for Google Chrome.

Features:-

- ⇒ written in C++
- ⇒ Uses Just-In-Time (JIT) compilation.
- ⇒ Converts JavaScript directly into machine code.
- ⇒ Used in Chrome and Node.js

10) How does JavaScript executes code?

1. Code is loaded

2. Parsing - Syntax is checked

3. Compilation - JIT compiler converts
Code into machine code.

4. Execution context creation:

- * memory allocation

- * variable and function hoisting

5. Execution phase

- * code runs line by line

6. Call Stack

- * keeps track of function execution.

7. Event loop

- * handles

asynchronous operation